

Citizen Identity Management Schema Documentation

The schema models real civil registration processes. It describes all tables, their columns, data types, constraints (primary keys, foreign keys, unique and check constraints), index details, and user roles.

Tables:

Person

The Person table is the main identity table of the system. It stores basic personal information for every individual, no matter their role. The primary key is `cnic` (VARCHAR(13)), which uniquely identifies each person. Other attributes are `first_name` and `last_name` (VARCHAR(50), NOT NULL), `gender` (ENUM: male, female, other), and `date_of_birth` (DATE, NOT NULL). The `created_at` field stores when the record was created and defaults to the current timestamp. This table acts as the parent for Citizen and Officer.

Citizen

The Citizen table represents legally recognized citizens who can apply for identity documents. It uses `cnic` as both its primary key and a foreign key referencing `Person(cnic)`, creating a one-to-one relationship. The `date_of_death` field (DATE, NULL) allows keeping historical records. The `created_at` field stores the creation time.

Parents

The Parents table stores biological parent-child relationships. It has an auto-increment `id` as the primary key. The `child_cnic` field is a foreign key referencing `Person(cnic)` and is unique, so one child has only one parent record. `father_cnic` and `mother_cnic` are nullable foreign keys referencing `Person(cnic)`. A CHECK constraint ensures a child cannot be their own parent.

Marriage

The Marriage table stores legal marriage records. The primary key is `marriage_id` (INT, AUTO_INCREMENT). `husband_cnic` and `wife_cnic` are foreign keys referencing `Person(cnic)`. `marriage_date` is required, and `divorce_date` is optional to support divorce cases. A CHECK constraint prevents self-marriage. A composite unique constraint prevents duplicate marriage records with the same divorce status.

Province

The Province table stores province information. `province_id` is the primary key, and `province_name` is unique and not null. This prevents duplicate province names and supports proper normalization.

City

The City table stores cities within provinces. `city_id` is the primary key. `city_name` (VARCHAR(100), NOT NULL) stores the city name. `province_id` is a foreign key referencing `Province(province_id)`. A composite UNIQUE constraint on (`city_name`, `province_id`) ensures no duplicate city names in the same province.

Address

The Address table stores physical addresses that can be shared by citizens, officers, and branches. `address_id` is the primary key. `building_no` and `street_no` are required. `city_id` is a

foreign key referencing City(city_id). landmark and postal_code are optional. This design avoids duplicate addresses and allows reuse.

Branch

The Branch table represents government service offices. branch_id is the primary key. It includes a unique branch_code, branch_name, and branch_type (ENUM: regional, city, mobile). address_id is a foreign key referencing Address(address_id). Other fields include contact information, operating hours, is_active status, and created_at timestamp.

Role

The Role table defines job roles for officers. role_id is the primary key. role_name is unique and not null. This supports role-based access control and avoids hard-coded roles.

Officer

The Officer table stores government employee data. The primary key is cnic, which is also a foreign key referencing Person(cnic). employee_id and email are unique. It includes foreign keys to Address (address_id), Branch (branch_id), and Role (role_id). Other fields are joining_date, is_active, password_hash, and created_at. This separates employment data from personal identity data.

Citizen_Address

The Citizen_Address table tracks address history of citizens. It has an auto-increment id as the primary key. citizen_cnic is a foreign key referencing Citizen(cnic), and address_id references Address(address_id). valid_from and valid_to store the time period when the address was valid. This allows storing past addresses.

Application

The Application table stores requests submitted by citizens for identity services. application_id is the primary key. citizen_cnic is a foreign key referencing Citizen(cnic), and assigned_officer_cnic references Officer(cnic). It includes application_type (ENUM), status (ENUM, default 'pending'), and timestamps for submission and processing. This is the main workflow table.

Payment

The Payment table stores payment records related to applications. payment_id is the primary key. application_id is a foreign key referencing Application(application_id) and is unique, allowing only one payment per application. Other fields are amount, payment_method, payment_status, paid_at, and received_by_cnic (foreign key to Officer).

ID_Card

The ID_Card table stores issued identity cards. id_card_id is the primary key. card_number is unique. application_id is a unique foreign key referencing Application(application_id). Other fields are issue_date, expiry_date, version_number, is_active, and created_at. An index on is_active helps quickly find active cards.

B_Form

The B_Form table stores birth registration documents. b_form_id is the primary key. application_id is a unique foreign key referencing Application(application_id). child_cnic is a unique foreign key referencing Citizen(cnic). Other fields include birth_registration_number, place_of_birth, and created_at.

Index Specifications and Rationale:

Primary keys are automatically indexed for fast lookup. Unique constraints also create indexes to protect data integrity on fields like CNIC, branch_code, card_number, and registration numbers. The index on ID_Card(is_active) improves performance for searching active cards. Composite unique constraints prevent logical duplicates in relationship tables.

User Roles Definitions:

Data Entry Operator

Responsible for entering citizen data and creating applications. This role has read and write access to basic citizen and application data but cannot approve or reject applications.

Supervisor

Responsible for reviewing and processing applications. Can approve or reject applications and supervise data entry operators.

System Administrator

Has full system access. Manages officers, roles, branches, system configuration, and audits.

Citizen

Has access to basic information like their own data and limited data of relatives. Access is view-only.